GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Cloud Automation

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TECHNICAL SKILLS

Languages & Core: TypeScript, JavaScript (ES6+), Python, Go, SQL (PostgreSQL, MariaDB), HTML5, CSS3, Linux/Bash

Frontend & Spatial Web: React 19, Next.js 15 (App Router, Server Components), Tailwind CSS 4, Vite 6, Three.js, WebGL2, Canvas API, Framer Motion, Lenis Scroll, Core Web Vitals (LCP, CLS < 0.02, FCP)

Backend & Distributed Systems: Node.js, Express, FastAPI, PostgreSQL, Supabase, MariaDB, Redis, BullMQ, WebSockets, REST APIs, Docker

Cloud, AI & Infrastructure: GitHub Actions 24/7 CI/CD Fleets, Meta Cloud API (WABA), Gemini AI Multimodal, FFmpeg, Railway, AWS (S3, Lambda), Exotel Telephony, Silero VAD, Sarvam AI, Groq LPU

KEY PRODUCTION PLATFORMS & SYSTEMS

Spatial & Creative Web Suite — SOLUM Minerals, Dream Heights & Branders **2025 – Present** | Open Source

Next.js 15, Three.js WebGL2, React 18, Vite 6, Canvas API, Framer Motion, Lenis Smooth Scroll *Production Platforms*

- **Procedural In-Memory Asset Synthesis:** Engineered procedural Three.js asset pipeline synthesizing dynamic 1024x1024 noise textures in-memory via Canvas API (SOLUM Minerals), eliminating 15MB static 3D mesh downloads and cutting initial bundle payload by 85%.
- **Hardware-Accelerated Physics & Web Vitals:** Decoupled Three.js render loops from DOM scroll events using GPU composite layers (transform3d) and Lenis virtualized interpolation, maintaining stable 60 FPS across 4K viewports with CLS < 0.01 and 99+ Lighthouse performance.

EduCore — Enterprise College Management ERP & Student Information System (SIS) **2025 – Present** | Open Source

Full-Stack Higher-Ed Campus ERP • React 19, TypeScript, Express, PostgreSQL, MariaDB Pools, Railway *Campus Platform*

- **Campus Administrative & Academic Management:** Architected comprehensive higher-education ERP managing end-to-end student lifecycles, admissions, department-level fee reconciliation, and academic grading across multi-tenant campus faculties.
- **Dual-Engine Database Layer & RBAC:** Engineered resilient database abstraction supporting pooled PostgreSQL and MariaDB instances with connection-drain health recovery (<15ms p90 query latency), paired with declarative 9-role JWT access controls and immutable audit logging.

yt-auto — 24/7 Autonomous Cloud Media Production Fleet **2025 – Present** | Open Source

Cloud Distributed Video Fleet • Python, FFmpeg, GitHub Actions CI/CD Matrix, Gemini Multimodal AI *Distributed Automation*

- **Event-Driven Cloud Processing Fleet:** Architected media processing fleet running multi-channel batch rendering across parallelized GitHub Actions matrix workers, orchestrating 150+ automated video builds weekly with zero dedicated server overhead.
- **Multimodal QA Gate & Filtergraph Mastering:** Automated video synthesis and audio mastering via custom FFmpeg filtergraphs with sidechain compression, gated by a Gemini Multimodal AI evaluation pipeline rejecting visual artifacts (≥85 score threshold).

Fasal Doctor — Autonomous Edge-AI Foliar Pathology & Crop Advisory Platform **2026 – Present** | Mobile System

Edge Computer Vision • Flutter, Dart, TFLite Edge, INT8 Post-Training Quantization, Worker Isolates *Agritech Platform*

- **Sub-50ms On-Device Neural Inference:** Deployed post-training INT8 quantized MobileNet foliar pathology neural networks running 100% offline on low-power mobile devices in rural connectivity dead zones, achieving sub-50ms inference and >94% diagnostic accuracy.
- **Multi-Isolate Architecture & Supply-Chain Auth:** Architected asynchronous Flutter worker isolates decoupling 60 FPS camera previews from tensor processing, integrated with micro-precipitation spray gating and cryptographic agrochemical barcode authentication.

WORK EXPERIENCE

CP Systems (Cool & Pure System) **2024 – 2025**

Software Developer • Industrial Water Purification & Dairy Machinery (cpsystem.in) *Sri Muktsar Sahib, Punjab, India*

- **Industrial Telemetry & Database Profiling:** Built internal operational telemetry, maintenance dispatch, and inventory microservices for a regional manufacturer of industrial RO plants and Bulk Milk Coolers (BMC) using Node.js, TypeScript, and SQL, profiling query plans to reduce p95 latency by 42%.
- **WhatsApp Business Automation & CI/CD:** Engineered automated customer communications and lead qualification over Meta Cloud API with 24-hour follow-up radars, while implementing GitHub Actions CI/CD pipelines with composite caching (reducing build validation times from 14m to 3.8m).

Gurdharam Enterprise **2025 – Present**

Full-Stack Software Engineer • Enterprise Software Consultancy (Govt. MSME: UDYAM-PB-03-0079744) *Punjab, India • Remote*

- **Commercial Client Delivery & WebGL Platforms:** Architected production web platforms, agency portals (Branders), and interactive 3D WebGL client applications, delivering consistent sub-800ms LCP and 0.00 CLS across multi-region edge CDNs.
- **Real-Time Conversational Telephony AI:** Built sub-700ms end-to-end conversational voice pipeline over Exotel full-duplex WebSockets, Silero VAD barge-in interruption, and streaming Sarvam AI Indic LLMs, alongside resilient Meta Cloud API webhook gateways.

EDUCATION & CREDENTIALS

Baba Farid Group of Institutions (BFGI) **2024 – Present**

Bachelor of Science (Hons.) • Applied Computational Systems & Computer Science Foundations *Punjab, India*

Senior Secondary Education (Class XII / Grade 12) — 91.0% (Academic Distinction) **2024**

Focus: Physics, Chemistry, Mathematics (PCM) • Quantitative & Algorithmic Foundations *Punjab, India*

Credentials & Verification: Meta Verified Business Entity • Official WhatsApp Business Platform Developer • Sarvam AI Startup Partner • Govt. MSME Registered Enterprise (UDYAM-PB-03-0079744) • Additional Systems: *SnapAR Studio (Camera Kit WebGL2)*, *TakeMyInterview AI (Local Voice LLM)*, *DoodHisaab (Mobile Dairy Ledger)*.